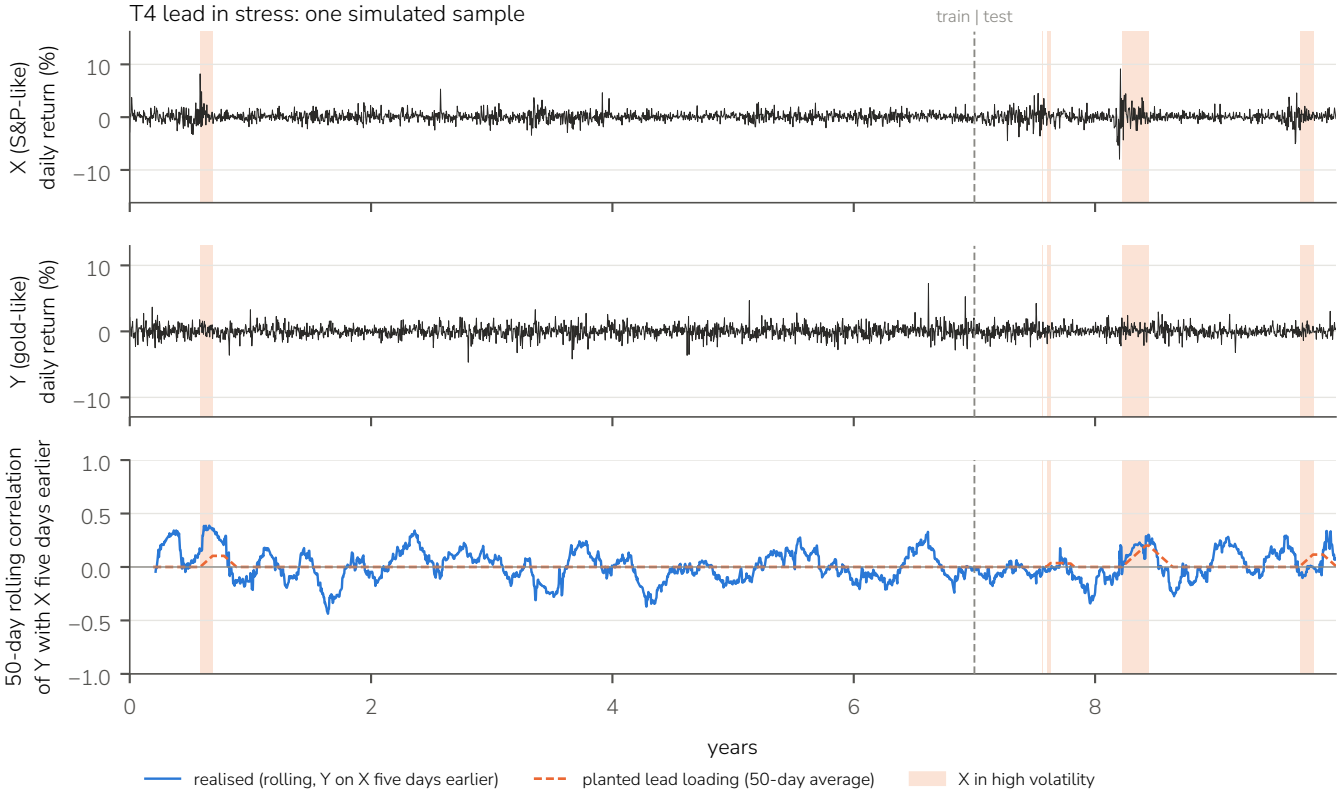


One replication of the simulation design, 10 years (2,520 days), calibrated on the S&P 500 (X) and gold (Y).
Planted phenomenon: correlation 0.3; when X is in high volatility, Y also loads 0.2 on X five days earlier.
States from the 50-day window ending the previous day. Dashed vertical line: 70/30 train/test split. Days beyond the plotted range are marked at the edge.